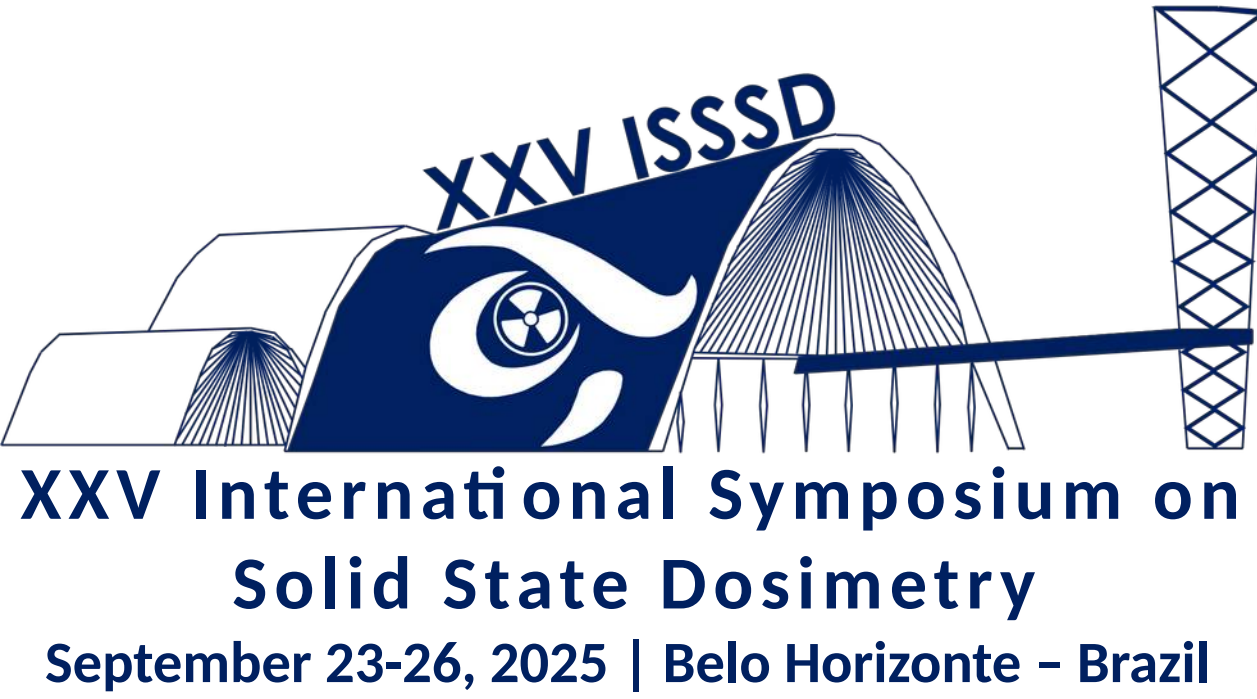


Characterization of the Ubatuba Formation: Integration of Gamma Radiometry, Total Organic Carbon, and Well Logs from the Upper Cretaceous to Lower Paleogene Intervals of the Campos Basin

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INTRODUCTION

In the Campos Basin, the intervals corresponding to the Cenomanian-Turonian and Cretaceous-Paleogene boundaries are characterized by the presence of rollover anticlines and listric faults, indicating a phase of intense tectonic activity during sediment deposition, strongly influenced by halokinesis. This tectonic regime played a decisive role in the structural organization and depositional patterns, driving significant environmental variations over time (Chang et al., 2006). This study involved the visual description and core orientation analysis of over 50 drilling cuttings samples from well 1-DEV-16-RJS, located in deep waters in the southern portion of the Campos Basin. The samples are characterized by well-spaced bathymetric levels, indicating a deep-water drilling context.

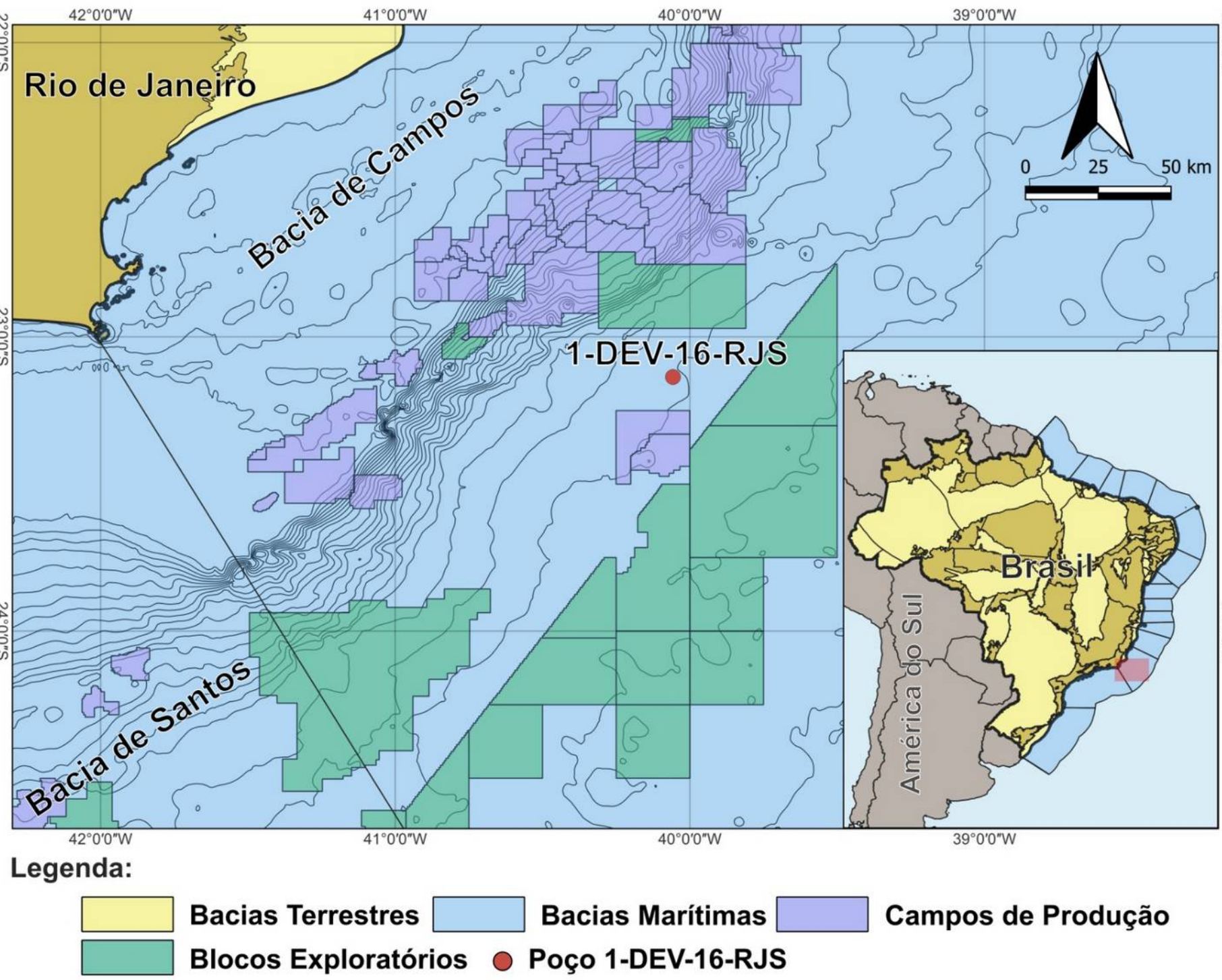


Figure1. Well Location Map.

MATERIALS AND METHODS

The nearest neighbor method relies on the inverse distance between points, along with the numerical value of the subspace where data is known. This way, drilling cuttings data can be brought to the same resolution as the well log curves.

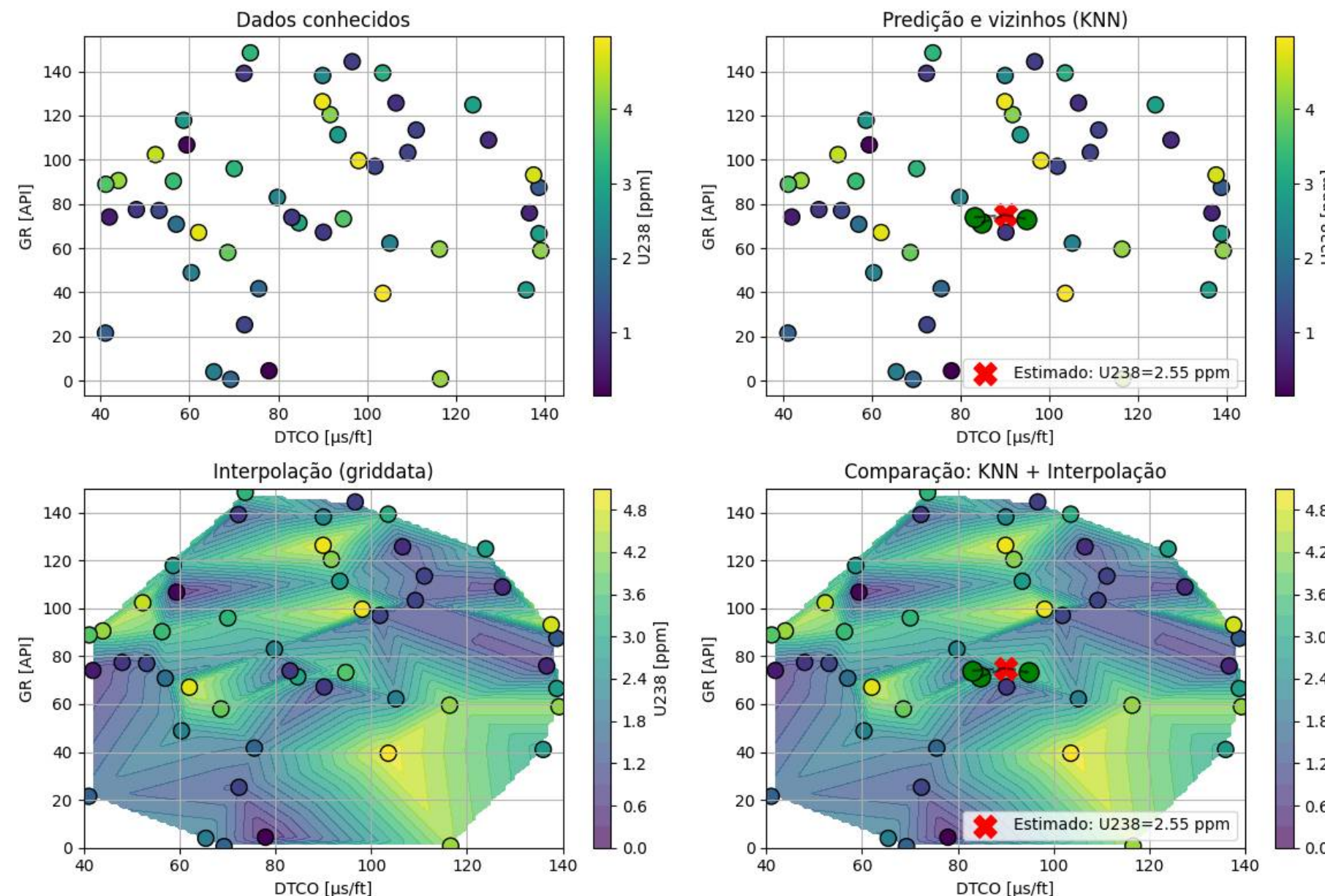


Figure2. KNN Regressor.

The execution of the Total Organic Carbon (TOC) tests, combined with the analyses obtained through HPGe, enabled the creation of laboratory test profiles using the KNN Regressor.

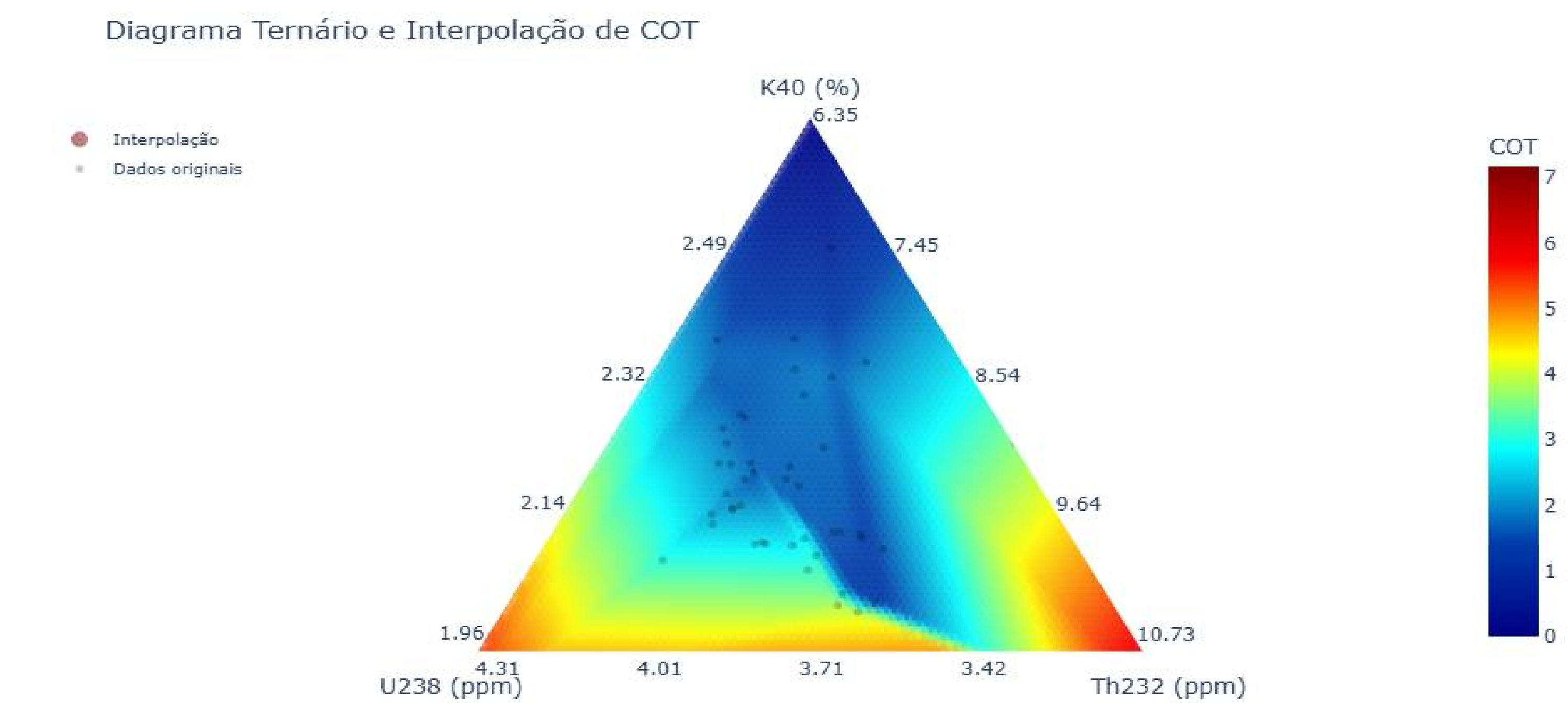


Figure3. Behavior of Total Organic Carbon in relation to radioisotopes.

RESULTS

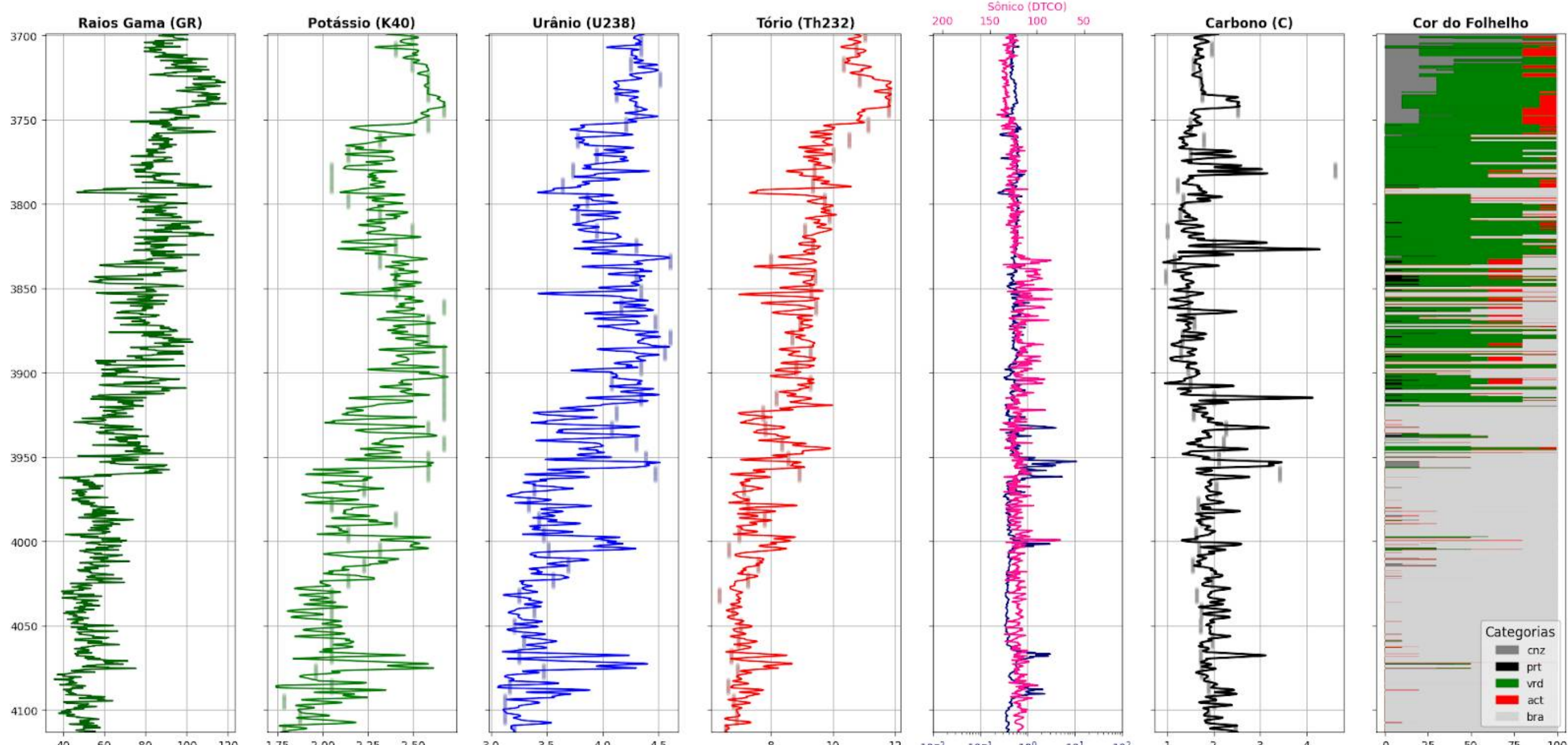


Figure4. Well GR, K40, eU238, eTh232, DTG with RT, TOC, and shale color description.

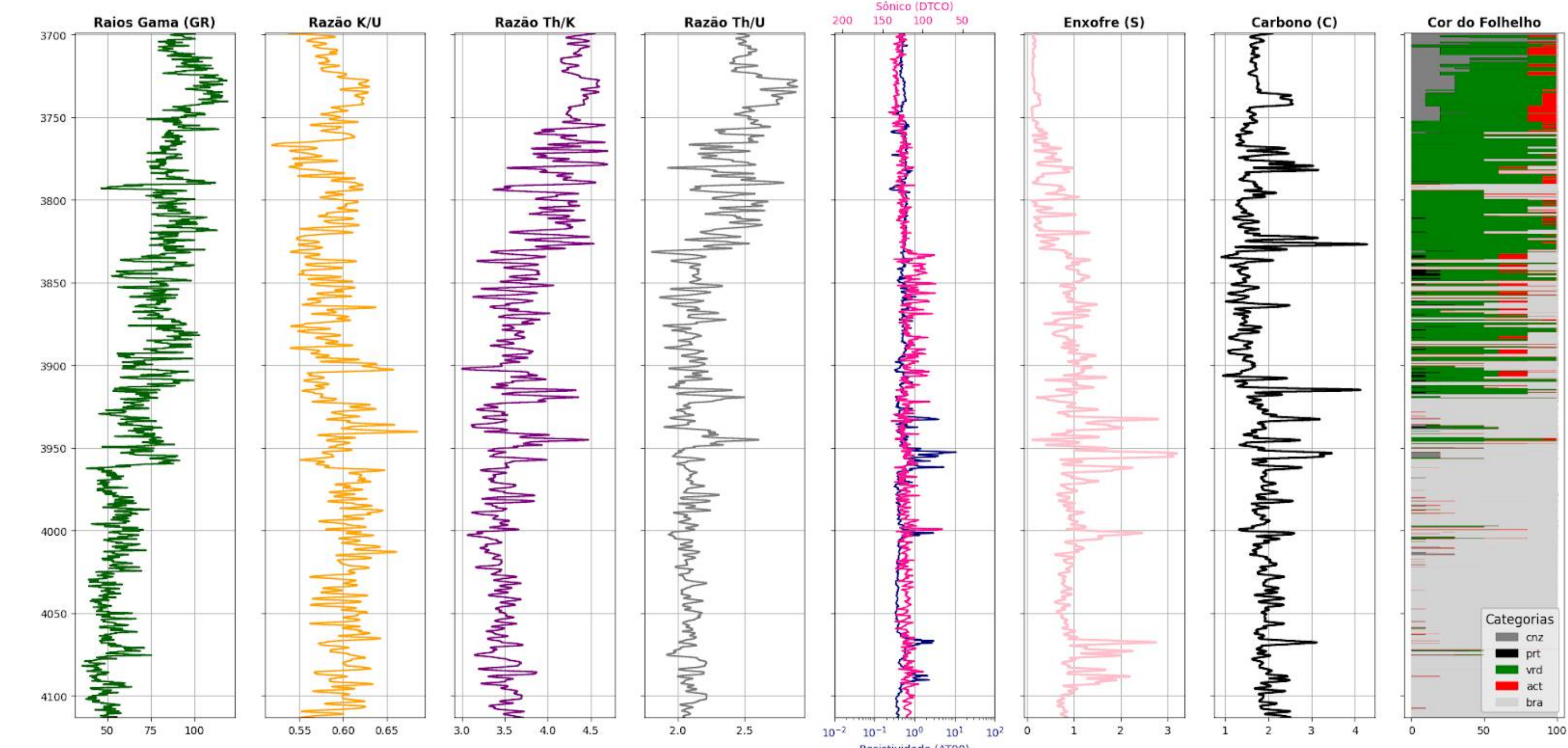


Figure5. Tracks for ratios of radioelements and sulfur log added.

DISCUSSION AND CONCLUSIONS

The eTh232/K40 ratio highlighted the transition between the "white" shale (bra) and the "green" shale (vrd), evidenced by an increase in terrigenous input, marked by higher eTh232 levels. The eU238 values indicate variations in water column thickness/accommodation space. The highest absolute Total Organic Carbon (TOC) contents are concentrated in the greenish shales. Elevated TOC concentrations are associated not only with eU238 levels but also with eTh232, suggesting a possible continental origin for the TOC increase in the study area.

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CHANG, H. K. *et al.* Interpretação e Mapeamento dos Sistemas Petrolíferos da Bacia de Campos. Tomo I. São Paulo: ANP/LEBAC/UNESP, 2006.
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